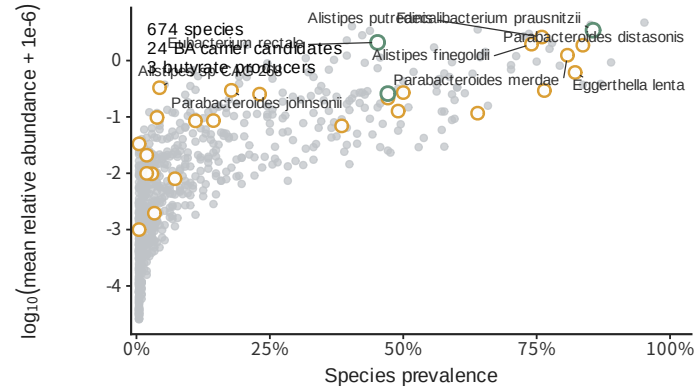


Extended Data Figure 1 | MetaPhlAn and HUMAnN profiling overview before downstream

MetaPhlAn and HUMAnN profiling provided a structured taxonomic, functional, coverage and species-attribution landscape suitable for downstream

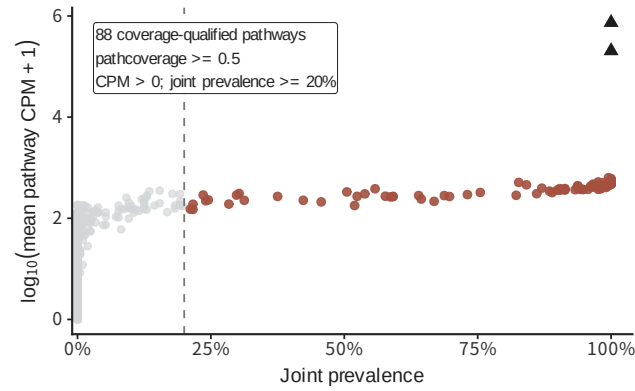
A MetaPhlAn species profile

Prevalence-abundance landscape across all locked samples



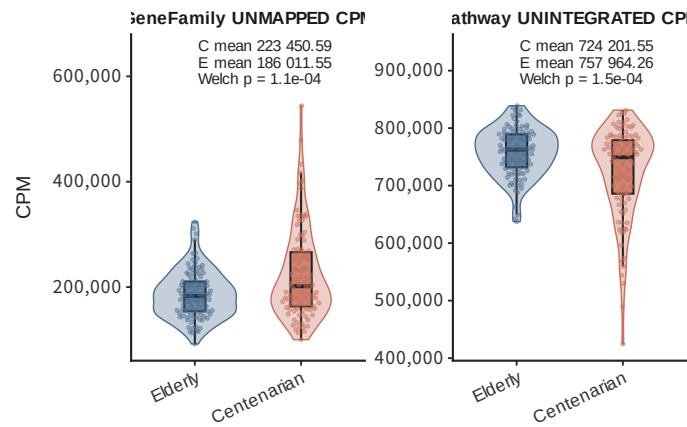
B HUMAnN pathway profile

MetaCyc pathway prevalence-abundance landscape



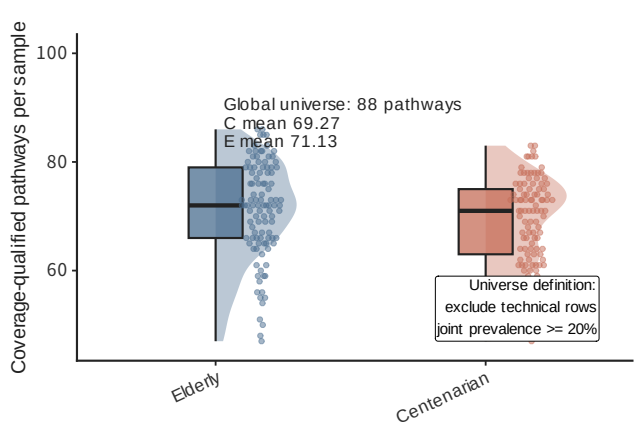
C HUMAnN technical annotation metrics

Annotation and mapping profile before downstream analysis

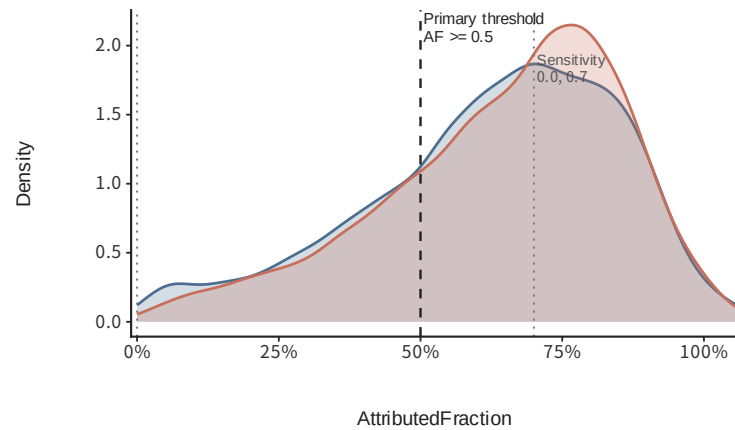


D Pathway coverage and qualified universe

pathcoverage >= 0.5, unstratified CPM > 0 within the global universe

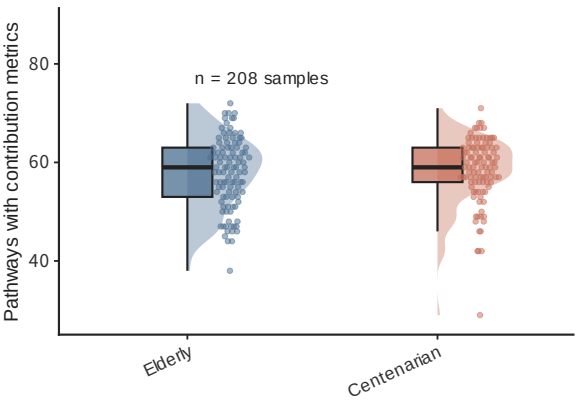


E1 AttributedFraction distribution



E2 Valid contribution metrics per sample

PrimaryAttributedFraction threshold >= 0.5



Elderly Centenarian Elderly Centenarian Elderly Centenarian